Real time Query Steering in Dense Retrieval using RL and Sparse Autoencoders (SAEs)

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Abstract

Dense retrieval models suffer from semantic drift (retrieving documents based on spurious distributional overlaps rather than contextual relevance), and correcting this typically requires expensive offline fine tuning. We propose Sparse Semantic Steering, a framework that corrects dense retrieval at inference time without retraining the base encoder. We project frozen embeddings through a Sparse Autoencoder (SAE) into a disentangled feature dictionary, then train a Proximal Policy Optimization (PPO) agent over a Context Aware State combining the dense query with top sparse features from a Pseudo Relevance Feedback (PRF) neighborhood. The agent learns bidirectional feature modifications (Negative Semantic Masking), projecting corrections back into the dense index. The policy is trained with simulated implicit feedback from relevance judgments but requires no labels at inference time. On five BEIR datasets the method achieves up to +61.7% rescue rate (NDCG@10) on ArguAna with negligible overhead (+4.30 MB VRAM, 0.002 ms routing).

CCS Concepts

• **Information systems** → **Retrieval models and ranking**; • **Computing methodologies** → *Neural networks*; **Reinforcement learning**.

Keywords

dense retrieval, reinforcement learning, sparse autoencoders, semantic steering, information retrieval

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1 Introduction

While dense retrieval has become the dominant approach in modern information retrieval, superseding the reliance on purely lexical matching by utilizing learned vector representations [13, 17], dense embeddings can struggle to distinguish fine grained semantic differences, making models vulnerable to hard negatives [36]: furthermore, they often fail to generalize across diverse domains, experiencing significant performance drops in zero shot settings due to domain shift [33]. Correcting this has historically required expensive offline fine tuning over hard negative distributions, a reactive strategy that scales poorly.

We propose leveraging simulated implicit feedback as an online correction mechanism. User interactions such as skipping top ranked results provide natural implicit preference feedback [15], and formulating this feedback as a reinforcement learning (RL) objective [1, 4] offers a principled way to steer queries dynamically. In this work we simulate feedback using benchmark relevance annotations (qrels); deployment with real click logs is a natural extension. Nonetheless, deploying RL directly over entangled dense spaces is intractable: the continuous action space lacks orthogonal basis directions, preventing agent convergence.

We resolve this by imposing a Sparse Autoencoder (SAE) bottleneck [2, 12] that projects dense embeddings into a disentangled feature dictionary. A lightweight Proximal Policy Optimization (PPO) agent [28] then operates on a Context Aware State combining the dense query with sparse Pseudo Relevance Feedback (PRF) features, learning bidirectional steering (Negative Semantic Masking) to suppress spurious features while a Rank Proportional Scaling Factor protects well performing queries.

Our contributions are:

(1) A sparse semantic steering architecture that transforms the intractable continuous dense perturbation problem into a constrained Markov Decision Process (MDP) over disentangled SAE features;
(2) A bidirectional action space with rank proportional torque that enables targeted feature suppression while protecting baseline retrieval quality;
(3) Empirical validation on five BEIR datasets showing consistent cross domain improvements (up to +61.7% rescue rate) with negligible overhead (+4.30 MB Video Random Access Memory (VRAM)).

2 Related Work

RL in Information Retrieval: RL has been applied to ranking [11], query reformulation [3, 22], and recommendation [1]. RLHF [4] has also been used to align language models [24, 30]. Existing RL agents in retrieval, by comparison, operate on discrete token spaces or candidate lists; training a continuous action policy directly on an entangled dense space $\mathbb{R}^d$ yields unstable gradients and sample inefficiency. Continuous vector perturbation in entangled spaces lacks an orthogonal basis for isolating semantic features, causing gradient updates along one direction to corrupt unrelated representational properties.

Sparse Autoencoders for Disentanglement: Dense embeddings suffer from superposition: networks pack more features than dimensions into almost orthogonal directions [5]. Early interpretability work analyzed individual neurons and circuits [23] or probed

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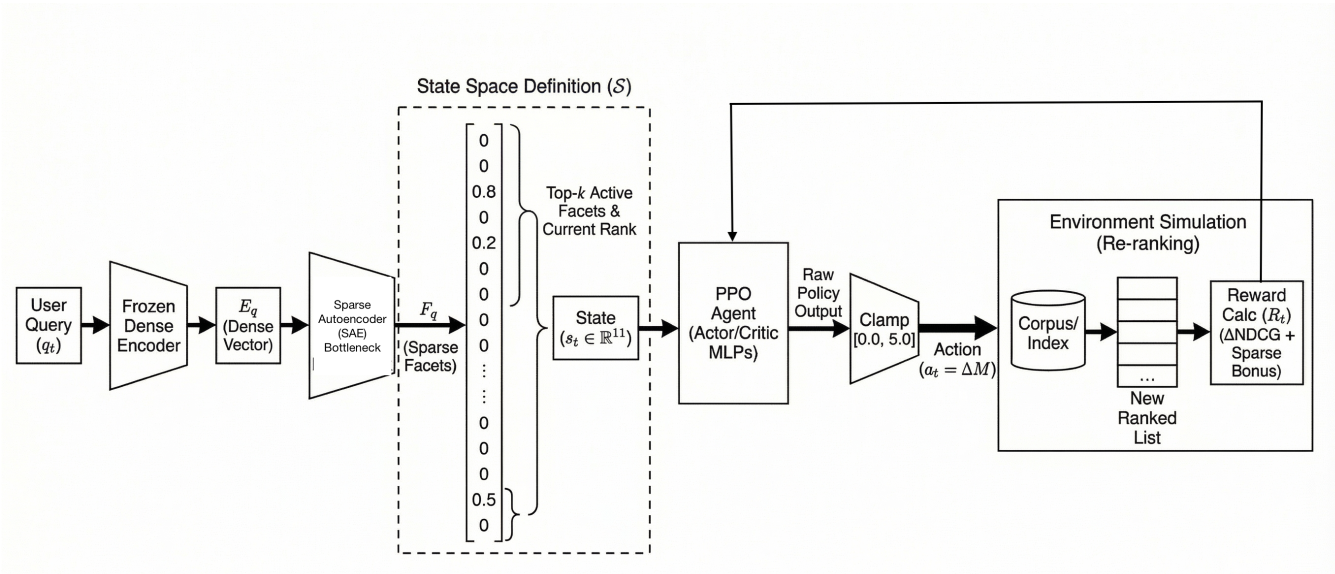

Figure 1: The Sparse Semantic Steering pipeline. A frozen dense encoder produces E_q, which is projected through an SAE bottleneck into sparse facets. The PPO agent receives a context aware state (top-k active facets from PRF), outputs a bidirectional steering action ΔM, and the corrected query is reranked against the corpus index. The reward is computed from rank changes during training only.

feed forward layers as key value memories [10], but these methods identify features without enabling targeted intervention on the entangled bases. SAEs address this by projecting activations into overcomplete dictionaries of monosemantic features [2, 12, 32]. We treat the SAE not as an analytical probe but as a structural tool: it transforms the dense embedding into a disentangled, manipulable sparse vector, converting the intractable continuous steering problem into a constrained MDP.

Controllable and Sparse Retrieval: Dense retrieval [13, 17] suffers from semantic drift [36], which is conventionally addressed through expensive hard negative retraining. Correcting the retrieval manifold by employing Approximate Nearest Neighbor (ANN) hard negative mining [35] is reactive: it shifts embeddings offline but provides no mechanism for inference time correction of individual queries. Sparse models like SPLADE [8] operate over the discrete lexical vocabulary [25], while late interaction architectures like ColBERT [18] recover exact match signals by preserving token level dense representations. Yet, neither approach allows for continuous amplification or suppression of abstract, multi token semantic concepts. Our framework bridges this gap by coupling an SAE with an RL agent for continuous semantic steerability.

3 Method

Standard dense retrieval scores relevance based on Maximum Inner Product Search (MIPS), $S(q,d) = E_Q(q) \cdot E_D(d)$ [16], but high dimensional embeddings suffer from anisotropy [7], and entangled dimensions cause semantic drift: irrelevant documents score above relevant ones. Solving for a direct adjustment $\Delta_q \in \mathbb{R}^d$ in the entangled space is ill posed, as arbitrary continuous shifts destroy unrelated semantic features. We bypass this by formulating correction as an episodic MDP [31] over a disentangled sparse basis. Figure 1 illustrates the full pipeline.

3.1 Sparse State Encoder

We project frozen dense embeddings $E \in \mathbb{R}^d$ into a sparse space $F \in \mathbb{R}^k$ $(k \gg d)$ using an L_1-penalized SAE: $F = \mathrm{ReLU}(W_{enc}E + b_{enc})$, $\hat{E} = W_{dec}F + b_{dec}$. The composite loss jointly enforces reconstruction fidelity, sparsity, and inner product preservation:

$$L_{SAE} = \|E - \hat{E}\|_2^2 + \lambda_1 \|F\|_1 + \lambda_2 \sum_{i,j \in \mathcal{B}} (E_i \cdot E_j - \hat{E}_i \cdot \hat{E}_j)^2 \quad (1)$$

where λ_1 = 1e-4 controls sparsity and λ_2 = 1e-3 preserves the MIPS ranking topology.

3.2 MDP Formulation

Operating RL on the full 4096-D SAE output would reintroduce dimensionality issues [20]. We instead extract a compressed, dynamic action space. The top k=10 retrieved documents are projected through the SAE, and a localized expansion vector is constructed:

$$F_{exp} = F_q + \sum_{i=1}^{k} w_i F_{d_i} \quad (2)$$

where weights w_i decay linearly from 1.0 to 0.1 by rank, normalized to $\sum w_i = 1$. The top 128 dimensions of F_{exp} define the active semantic facets for the agent.

State Space ($\mathcal{S}$): We define two configurations to separate training supervision from inference: **Training state** $s_t^{\text{train}} \in \mathbb{R}^{898}$: the

Table 1: Cross Domain Retrieval Performance (NDCG@10, inference time state). Bold = best per row.

Dataset	Domain	SPLADE	ColBERT	Dense Base	RL steered (Ours)	Rescue
SciDocs	Scientific	0.1395	0.1297	0.1143	**0.1506**	+36.7%
ArguAna	Argumentative	0.3588	0.3382	0.3371	**0.4518**	+61.7%
NFCorpus	Medical	0.3424	0.2923	0.2662	**0.3595**	+59.6%
FiQA	Financial	**0.4655**	0.4014	0.1931	0.2755	+29.1%
TREC COVID	Bio Medical	**0.9552**	0.9244	0.3611	0.4110	+87.5%

Table 2: Compute & VRAM Efficiency.

Architecture	Params	VRAM	Latency
ColBERT	~110 M	High	15–45 ms
Dense Base	~110 M	Base	0.001 ms
RL steered (Ours)	+ 6.56 M	+ 4.30 MB	0.002 ms

768-D dense query, a 128-D PRF sparse magnitude vector, a $\log_2$-normalized target rank scalar, and a $\log_2$ rank delta scalar. **Inference state** $s_t^{\text{infer}} \in \mathbb{R}^{896}$: only the dense query and PRF magnitudes, with no access to relevance labels or target document identity. All results in this paper use the inference time state exclusively.

Action Space ($\mathcal{A}$): The agent outputs $\Delta M \in [-2.0, 2.0]^{128}$, enabling bidirectional Negative Semantic Masking. The sparse action is projected back by the SAE decoder weights, scaled by a Rank Proportional torque:

$$\tau = \max\left(0.1, \min\left(1.0, \frac{\log_2(\rho_{init})}{\log_2(|C|)}\right)\right) \tag{3}$$

yielding the corrected query: $E'_q = E_q + \tau(\Delta M\, W_{dec}^T)$. Well ranked queries ($\rho_{init} \approx 1$) receive near zero correction; poorly ranked queries receive maximal adjustment.

Reward Function ($\mathcal{R}$): Following reward shaping principles [21], we construct a Logarithmic Rank Delta Reward from relevance judgments (qrels), simulating implicit feedback [15]:

$$R = 10.0 \times (\log_2(\text{Rank}_{old} + 1) - \log_2(\text{Rank}_{new} + 1)) \tag{4}$$

augmented with discrete bonuses (+20.0 for Top 10, +10.0 for Top 3). This requires a known relevant document during training only.

3.3 PPO Optimization

We optimize the steering policy using PPO [28], an actor critic algorithm [19] chosen for gradient stability. The Actor $\pi_\theta(a_t|s_t)$ proposes ΔM and the Critic $V_\phi(s_t)$ reduces variance by leveraging Generalized Advantage Estimation (GAE) [27]. While PPO's clipped surrogate objective ($\epsilon = 0.2$) is designed to bound policy steps, we rely on its specific code level optimizations to truly prevent destructive updates to the query geometry [6]. Full formulation details are in supplementary materials.

4 Experiments

To evaluate the empirical performance of Sparse Semantic Steering, we conduct a series of retrieval experiments across diverse domains. To support reproducibility, all code, data splits, and pretrained Sparse Autoencoder (SAE) weights are publicly available in our anonymized repository: https://anonymous.4open.science/r/IR-interpretability-RL-ABF2/.

4.1 Setup

We evaluate on five BEIR [33] subsets spanning scientific (SciDocs), argumentative (ArguAna), medical (NFCorpus), financial (FiQA), and biomedical (TREC COVID [34]) domains. The base dense encoder (Contriever) is frozen; the steering policy is trained per domain using target dataset relevance labels. We report NDCG@10 [14]. Statistical significance is determined by means of paired randomized permutation tests ($p < 0.05$) [29]. We define *rescue rate* as $(\text{NDCG}_{RL} - \text{NDCG}_{Dense})/(\text{NDCG}_{SPLADE} - \text{NDCG}_{Dense}) \times 100\%$.

We compare against three retrieval architectures as baselines: **Contriever** [13] (frozen dense), **SPLADE v2** [9] (learned sparse), and **ColBERTv2** [26] (late interaction). The SAE is pretrained for 100 epochs using the composite loss (Eq. 1); the PPO agent trains for 300 episodes. Architecture and hyperparameter details are in supplementary materials.

4.2 Results

Table 1 shows results across all five datasets. On abstract reasoning domains (SciDocs, ArguAna, NFCorpus), the steered agent achieves up to +61.7% rescue rate on ArguAna (0.4518 NDCG@10), outperforming both SPLADE and ColBERT. On jargon-heavy tasks (FiQA, TREC COVID), the method improves over the dense baseline (+87.5% rescue on TREC COVID) but SPLADE retains the absolute lead. The gap stems from vocabulary coverage: SPLADE's masked-language-model pre-training injects domain-specific lexical tokens (e.g., financial tickers, biomedical MeSH terms) that the frozen Contriever encoder never represented. Sparse steering cannot recover tokens absent from the base encoder's embedding space; it corrects semantic drift within the encoder's representational capacity but cannot synthesize missing vocabulary. Training dynamics (Figure 2) confirm the agent converges to a generalized correction strategy rather than memorizing individual queries. As shown in Table 2, the entire pipeline adds only +4.30 MB VRAM and 0.002 ms routing latency (PPO forward pass) over the frozen dense baseline, keeping the same index format and per-document footprint as standard single-vector retrieval.

Illustrative Example: On ArguAna, the query "Should performance enhancing drugs be accepted in sports?" ranked the target counter argument at position 112 under the dense baseline, which surfaced documents about general drug criminalization rather than

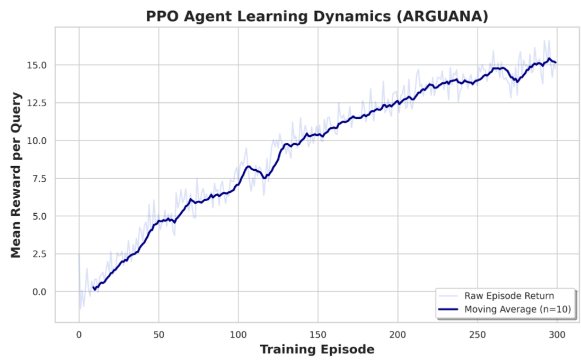

Figure 2: PPO agent reward (ArguAna, 300 episodes). The 10-episode moving average increases monotonically, confirming convergence to a generalized correction strategy rather than query memorization.

Table 3: Ablation Studies (ArguAna test split, inference time state).

Configuration	NDCG@10	Δ	Degrad.
Full Architecture (Ours)	0.4518	–	–
w/o Context Aware State	0.3522	-0.0996	-22.05%
w/o Negative Masking	0.4192	-0.0326	-7.22%
w/o Dynamic Torque	0.4248	-0.0270	-5.98%
Vanilla RL (no SAE)	0.3277	-0.1241	-27.47%

athletic regulation. The steering agent amplified sparse dimensions that, upon post hoc inspection, correspond to competitive fairness (+1.82) and athletic physiology (+1.45), while suppressing dimensions related to narcotics trafficking (−1.90) and criminal sentencing (−1.55), pushing the target from Rank 112 to Rank 2. These facet labels are assigned post hoc for illustration; systematic interpretability validation is deferred to future work.

4.3 Ablation

Table 3 isolates component contributions. Removing the Context Aware State causes the largest drop (−22.05%): without original query context, the agent applies generic corrections. Operating directly on the dense space without SAE projection (Vanilla RL) yields 0.3277, *below* the frozen Contriever baseline (0.3371), confirming that the disentangled sparse projection is necessary for effective RL-based steering. Negative Masking and Dynamic Torque each contribute meaningfully (−7.22% and −5.98%). The Vanilla RL result is particularly instructive: without the SAE bottleneck, the agent degrades retrieval below the unmodified baseline, indicating that continuous action RL in entangled dense spaces fails to converge to a useful policy.

5 Conclusion

We introduced Sparse Semantic Steering, a framework that corrects semantic drift in dense retrieval at inference time by projecting embeddings through an SAE bottleneck into a disentangled feature space and training a PPO agent to steer queries by applying bidirectional feature modifications. On five BEIR datasets, the method achieves up to +61.7% rescue rate on ArguAna, outperforming SPLADE and ColBERT at single-vector speed, with +4.30 MB VRAM and 0.002 ms routing overhead.

Deployment Considerations: Because the dense corpus index is never modified by policy updates, the 6.56M-parameter PPO agent can be retrained on fresh click telemetry and hot swapped at the query router without reindexing. In Retrieval Augmented Generation (RAG) pipelines, the steered retriever functions as a pre generation semantic filter: by suppressing drifted documents before they enter the language model's context window, it reduces downstream hallucination risk without additional inference cost.

Limitations: The reward signal uses benchmark relevance judgments rather than real user interactions; transfer to live click based feedback is untested. The baseline comparison does not include supervised or contextual bandit alternatives. Systematic interpretability evaluation of the sparse features is deferred to future work. The policy is trained per domain; cross domain generalization of a single policy has not been evaluated. Extended formulations, dataset statistics, hyperparameters, and additional analyses are available in the supplementary materials.

Statement of AI Use

The authors acknowledge the use of Gemini AI to refine the text, code, and generate the images included in this paper.

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